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Hierarchy-Feature Representation Model for Hierarchical Text Classification

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Abstract

Hierarchical text classification is an essential subtask of multi-label text classification with a taxonomic hierarchy. Existing approaches often cannot fully utilize hierarchical information, as they tend to neglect the semantic relationships between texts and labels. To address these limitations, I propose the Hierarchy Feature Representation Model (HiFRM), a novel framework designed to enhance the performance of the model by integrating hierarchical and text feature representations. The proposed HiFRM incorporates an end-to-end hierarchy-aware text feature propagation mechanism, which leverages fine-grained label-related information to improve the model's capability to capture complex label dependencies. Additionally, I introduce a global label feature-aware evaluation module that dynamically balances the importance of frequent and rare labels, ensuring a more comprehensive evaluation and improving model performance on imbalanced datasets.

My approach has been validated through extensive experiments on multiple benchmark datasets, including datasets with varying levels of label imbalance and hierarchy complexity. The experimental results demonstrate that HiFRM consistently outperforms the state-of-the-art methods with respect to Micro F1 and Macro F1 scores. In particular, the model significantly improves the classification of rare labels, addressing one of the most critical challenges in HTC. Ablation studies further confirm the contributions of the proposed hierarchy-aware propagation mechanism and the feature-aware evaluation module to the overall performance. These results highlight the effectiveness and robustness of HiFRM in tackling the challenges of hierarchical text classification.

Keywords: *Hierarchical Text Classification, Text Feature Propagation, Label Feature Awareness, Weighted Evaluation Metrics*

Chapter 1

Introduction

Text classification is a fundamental task in Natural Language Processing (NLP), with wide-ranging applications such as sentiment analysis[9], information retrieval [7], and document categorization [16]. Hierarchical text classification (HTC) represents a specialized form of multi-label text classification, where classification outcomes are assigned to one or more nodes within a hierarchical label structure [12]. These hierarchical labels are typically represented as a tree or a directed acyclic graph, forming paths that progress from coarse-grained to fine-grained labels [17]. This hierarchical organization has proven particularly effective for tasks such as news categorization [6] and scientific paper classification [5].

Existing hierarchical text classification methods can generally be divided into two main categories: local and global approaches. Local methods typically involve the construction of multiple classification models and the traversal of the hierarchy in a top-down manner. Previous local studies[11, 14, 1] have sought to address data imbalance in the child nodes by incorporating parental node learning. In contrast, global methods frame HTC as a flat multi-label text classification problem, employing a single classifier for all categories. Recent global methods have introduced various strategies to harness structural information from top-down pathways, including recursive regularization[4], reinforcement learning[17], and meta-learning[15].

However, these local models often require many parameters, lack comprehensive structural information, and are susceptible to exposure bias, affecting their overall effectiveness. Although global methods aim to streamline the model by employing a single classifier for all categories, they also face limitations. Currently, no global method effectively integrates text-related features into a unified label structure or evaluates

model performance in a manner that fully considers the relationships and dependencies among global label features.

To address the limitations above, This paper proposes the Hierarchy-Feature Representation Model (HiFRM), a hierarchy-aware framework for hierarchical text classification that utilizes end-to-end text feature propagation and incorporates a global label feature evaluation module to enable a more detailed performance assessment. The hierarchy is represented as a directed graph in this framework, and a hierarchy-aware structural encoder is introduced to model label dependencies. To evaluate the model's performance after label flattening, the quantitative label characteristics are captured on the basis of prior label decision tree information, and a specific weighting matrix is used to assess whether the model's performance is balanced between common and rare labels. Using the hierarchical encoder, HiFRM directly inputs text features and employs a text feature propagation model in a deductive manner. Empirical results demonstrate that this model achieves substantial performance improvements across multiple benchmark datasets, particularly notable gains on low-frequency labels. The proposed model effectively addresses the challenges of data imbalance and exposure bias by integrating global label features with hierarchy-aware text representations. In summary, the significant contributions of this paper are as follows.

1. Proposal of an end-to-end hierarchy-aware text feature propagation model: This method improves performance by effectively leveraging fine-grained label-related information within the hierarchical structure.

2. Development of a global label feature-aware evaluation module: This module enhances the model's ability to address challenges associated with rare labels, improving overall performance.

3. Comprehensive experimental validation: Extensive experiments on multiple benchmark datasets demonstrate that the proposed model outperforms existing methods.

Chapter 2

Related Work

2.1 Hierarchical Text Classification

Hierarchical text classification is a task within multi-label classification, where the goal is to assign classification results to one or more nodes within a hierarchical structure. Existing state-of-the-art methods often encode hierarchical constraints from a global perspective, such as using directed graphs or tree structures. For example, the Hierarchy-Aware Global Model[19] proposed a hierarchy-aware global model to exploit the prior probability of label dependencies. Similarly, the multi-graph aggregation model[8] introduced three types of label knowledge graphs, including taxonomy graphs, semantic similarity graphs, and co-occurrence graphs, to enhance hierarchical text classification.

However, these methods have limitations as they cannot consider the interaction between the semantics of the label and the text, making it challenging to capture complex dependencies of the label and achieve comparable classification scores. of the label[13]. Consequently, their performance remains restricted[2].

Recent studies have focused on leveraging joint embeddings of text and labels to address these issues and establish hierarchy-aware matching mechanisms. This allows capturing the relationships among the semantics of the labels and distinguishing the correct labels from the incorrect ones within the hierarchy[3]. These advancements inspire further optimization of HTC models to handle complex hierarchical dependencies and improve classification performance.

2.2 Text Encoder

Text encoders are crucial in hierarchical text classification tasks, extracting meaningful features from input sequences. Traditional methods commonly employ recurrent neural networks (RNNs), such as long- and short-term memory (LSTM) networks and gated recurrent units (GRUs) or convolutional neural networks (CNNs) to capture contextual features. For example, bidirectional GRUs have been used with k-top max pooling to extract contextual features and generate salient n-gram representations[19]. To model global dependencies within the label hierarchy, some approaches integrate graph convolutional networks (GCNs) to encode label relationships and combine them with text features, thereby creating hierarchy-aware representations.

Recent advances highlight the importance of incorporating hierarchical prior information. Techniques such as encoding the label hierarchy as directed graphs, using label dependency graphs, or leveraging co-occurrence graphs have proven effective. Mapping text features into node spaces explicitly enables models to utilize hierarchical paths through GCNs, enhancing classification performance. Additionally, the architecture of text encoders often includes linear transformations and activation functions, such as ReLU, to optimize representation learning. This design facilitates the exploration of complex semantic interactions, further boosting the effectiveness of hierarchical text classification.

2.3 Evaluation Matric

In hierarchical text classification tasks, the choice of evaluation metrics is critical for assessing model performance. Traditional evaluation methods typically use standard multilabel classification metrics such as precision, recall, micro F1 and macro F1[18]. Micro F1 emphasizes frequent labels, while Macro F1 assigns equal weight to all labels, highlighting performance on rare labels.

Based on these methods, feature-aware evaluation metrics have been developed to further enhance hierarchical classification evaluation. By introducing specific weighting matrices, these metrics enable weighted scoring of labels based on their roles within the hierarchy. This approach improves the applicability of traditional metrics to hierarchical classification tasks and provides a more comprehensive assessment of model performance.

Chapter 3

Methodology

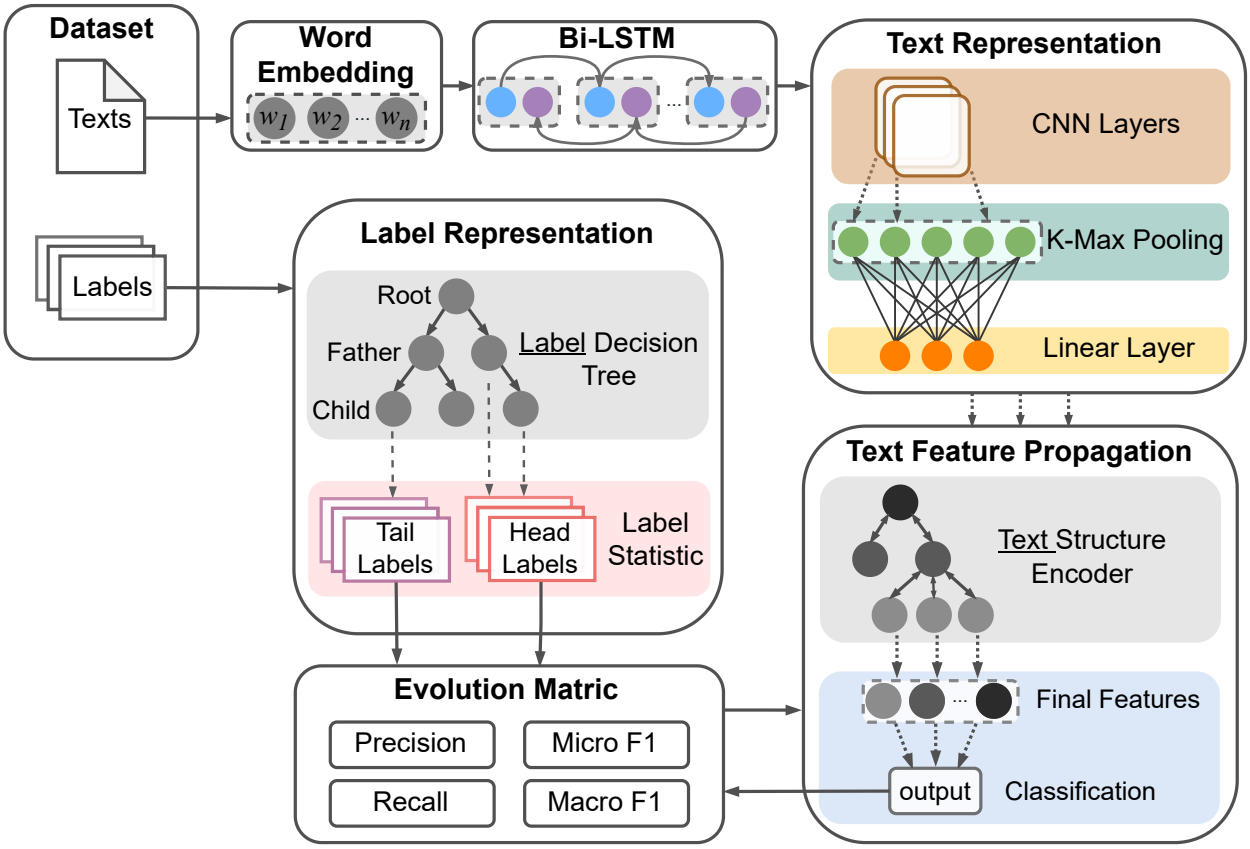


Fig. 3.1 The overall structure of Hierarchy-Feature Representation Model

3.1 Prior Hierarchy Information

3.1.1 Label Decision Tree

The hierarchy is predefined based on a holistic corpus. It organizes label subsets according to class relations, with the numbers representing the prior probabilities of label dependencies. I leverage the prior probabilities of label dependencies as prior hierarchical knowledge. Suppose that there exists a hierarchy path $e_{p,c}$ between the parent node v_p and the child node v_c. This edge feature $f(e_{p,c})$ is represented by the prior probability $P(U_c|U_p)$ and $P(U_p|U_c)$ based on the training corpus as follows:

$$P(U_c|U_p) = \frac{P(U_c \cap U_p)}{P(U_p)} = \frac{P(U_c)}{P(U_p)} = \frac{N_c}{N_p},$$

$$P(U_p|U_c) = \frac{P(U_p \cap U_c)}{P(U_c)} = \frac{P(U_p)}{P(U_c)} = 1.0,$$

where U_k denotes the occurrence of v_k and $P(U_c|U_p)$ is the conditional probability of v_c given that v_p occurs. $P(U_c \cap U_p)$ is the probability of $\{v_c, v_p\}$ occurring simultaneously. N_k represents the number of occurrences of U_k in the training subset. Note that the hierarchy ensures U_k given that $v_{\text{child}(k)}$ occurs. I rescale and normalize the prior probabilities of the child nodes $v_{\text{child}(k)}$ to ensure that their sum is equal to 1.

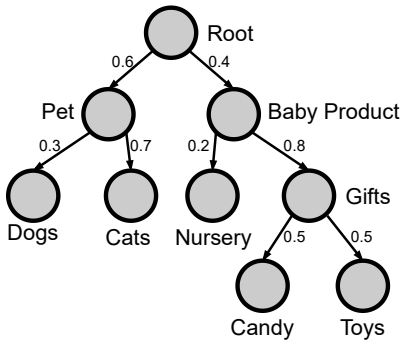


Fig. 3.2 Example of the taxonomic hierarchy.

3.1.2 Hierarchical Weight and Probability Representation

In hierarchy information preprocessing, this model defines a hierarchical structure comprising father and child nodes with associated weights and probabilities. In the following equations, the weight vector W represents the hierarchical nodes. Specifically,

W_f refers to the weight vector for the father nodes, with dimensions $N_{\text{father}} \times 1$, while W_c represents the weight matrix for the child nodes, with dimensions $N_{\text{father}} \times N_{\text{child}}$. Each row of W_c corresponds to the weights of the child nodes associated with a specific father node. The probability vector P captures the occurrence probabilities and corresponds to both W_f and W_c.

Weight Matrix Formula Calculate the weighted probability of occurrence for the nodes, represented as:

$$S_f = W_f \odot P_f$$

$$S_c = W_c \odot P_c$$

S is the weighted occurrence probability vector for father and child nodes, with a dimension of $N_{father} \times 1$ and $N_{father} \times N_{child}$. To combine father and child nodes into a single vector, I define a global weighted probability matrix S:

$$S = \begin{bmatrix} S_f \\ S_c \end{bmatrix}$$

Head and Tail Sets After a global weighted probability matrix has been built, the model sorts the nodes in set S by their weighted probabilities in descending order. Then, select the top k nodes from the sorted list to form the head set, which represents the nodes with the highest occurrence probabilities. Similarly, select the bottom k nodes from the sorted list to form the tail set, representing the nodes with the lowest occurrence probabilities.

3.2 Text Representation

Previous global models classify labels upon the original textual information and improve the decoder with predefined hierarchy paths. In contrast, I construct a novel end-to-end hierarchy-feature representation model (HiFRM) for the mutual interaction of text features and label correlations. It combines a traditional text classification model with a hierarchy encoder, thus obtaining label-wise text features.

Word Embedding Given a document $x = (w_1, w_2, ..., w_s)$, each word w_i is first converted into a vector representation using word embeddings.

Bi-LSTM These word embeddings are then input into the LSTM layer to capture contextual information. To induce label correlations, HiFRM employs a bidirectional

LSTM by the fusion of a child sum and a top-down module:

$$\tilde{h}_k^{\uparrow} = \sum_{j \in \text{child}(k)} f_p(e_{k,j}) h_j^{\uparrow},$$

$$\tilde{h}_k^{\downarrow} = f_c(e_{k,p}) h_p^{\downarrow},$$

$$h_k^{bi} = h_k^{\uparrow} \oplus h_k^{\downarrow},$$

where $h_k^{\uparrow}$ and $h_k^{\downarrow}$ are separately calculated in the bottom-up and top-down manner as $h_k = \text{BiLSTM}(\tilde{h}_k)$. $\oplus$ indicates the concatenation of hidden states. The final hidden state of node k is the hierarchical node representation h_k^{bi}.

CNN Layers Multiple convolutional neural networks (CNNs) are applied to extract n-gram features, with each CNN layer responsible for capturing information from specific n-grams.

Top-k Max Pooling After concatenating all $n-gram$ features, a $top-k$ max-pooling layer is applied to extract the top k maximum values, thereby selecting key features and retaining the most important information.

Linear Layer In the linear layer, the model inputs the text features S into a linear layer for transformation, giving the node inputs V. The calculation of V is based on the formula $V = M \times S$, where:

- $S \in \mathbb{R}^{n \times d_s}$ represents the matrix of input text features, with $n = n_k \times n_c$ represents the product of the top-k number and d_s indicating the dimension of each feature.
- $M \in \mathbb{R}^{(n \times d_c) \times (C \times d_v)}$ is a trainable weight matrix used to project the text features from their original space to the input space of the node.
- $V \in \mathbb{R}^{C \times d_v}$ denotes the inputs of transformed nodes, where C is the number of categories in the hierarchy, and d_v is the dimension of the representation of each node.

3.3 Text Feature Propagation

Structure Encoder The input nodes $V$are further processed by the structure encoder, which features a message-passing mechanism based on Hierarchy-GCN. This mechanism propagates and updates information between nodes according to a predefined hierarchical structure, allowing the model to capture global relationships within the

hierarchy. Within the structure encoder, each node receives information from neighboring nodes and updates its representation through message passing. This updated node representation incorporates information from both its own hierarchical level and connected levels, resulting in a richer hierarchical representation.

Progressive Feature Aggregation During the iterative update process in the structure encoder, the model progressively passes and aggregates text feature information layer by layer, ultimately obtaining a hierarchy-aware representation for each node. This representation encapsulates textual information from various hierarchical levels, enabling the model to express hierarchical text features effectively.

3.4 Classification

After completing structure encoding, the final node representation is used as input to the classifier to produce the classification result. During training and inference, the model leverages the node representation from the structure encoder to perform classification, outputting the corresponding label or prediction.

I flatten the hierarchy by taking all nodes as leaf nodes for multi-label classification, no matter it is a leaf node or an internal node. The final hierarchy-aware features are fed into a fully connected layer for prediction. HiFRM is complementary with recursive regularization[4] as $L_r = \sum_{i \in C} \sum_{j \in \text{child}(i)} \frac{1}{2} \|w_i - w_j\|^2$ for the parameters of the final fully connected layer. For multi-label classification, HiFRM uses a binary cross-entropy loss function: $L_c = -\sum_{i=1}^{N} \sum_{j=1}^{C} \left[y_{ij} \log(y'_{ij}) + (1 - y_{ij}) \log(1 - y'_{ij}) \right]$, where y_{ij} and y'_{ij} are the ground truth and sigmoid score for the j-th label of the i-th sample. Thus, the final loss function is $L_m = L_c + \lambda \cdot L_r$.

3.5 Feature-Aware Evaluation Matric

Before assigning individual nodes their specified weights, I use standard evaluation metrics [4] for basic evaluation.

- **Precision** is the ratio of correctly predicted positive samples to all predicted positive samples, defined as $\frac{\sum TP}{\sum (TP+FP)}$.
- **Recall** is the ratio of correctly predicted positive samples to all actual positive samples, given by $\frac{\sum TP}{\sum (TP+FN)}$.

- **Micro F1** considers overall precision and recall across all instances, giving more weight to frequent labels, and is defined as $\frac{2\times Precision\times Recall}{Precision+Recall}$.
- **Macro F1** is the average F1 score across all labels, assigning equal weight to each label, and is defined as $\frac{1}{N}\sum_{i=1}^{N} F1_i$, where each class-specific $F1_i$ can be calculated using the Precision and Recall of the class as $\frac{2\times Precision_i\times Recall_i}{Precision_i+Recall_i}$.

The feature-aware evaluation method uses a weighting matrix to assign weights to nodes based on their inclusion in the Head or Tail set, adjusting scores across labels with varying frequencies. This approach enables weighted scoring of nodes, effectively allowing the model to balance its evaluation across high-frequency and low-frequency labels. The specific steps are as follows:

Construction of the Weighting Matrix Define the weight W_i, where the value depends on whether the node is in the Head or Tail set:

$$W_i = \begin{cases} w_h, & \text{If node } i \in \text{Head} \\ w_t, & \text{If node } i \in \text{Tail} \\ 1, & \text{Otherwise} \end{cases}$$

Nodes in the Head set are assigned a lower weight w_h, as these nodes appear more frequently. Nodes in the Tail set are assigned a higher weight w_t due to their lower frequency. Nodes not classified as Head or Tail are given a weight of 1.

Calculation of the Scoring Matrix The scoring matrix S is calculated by multiplying the weighting matrix W_i with the base score F_1:

$$S = W_i \times F_1$$

Using this weight-based evaluation method, the model's overall score reflects its performance across labels with varying frequencies, offering a more comprehensive measure of its effectiveness.

Chapter 4

Experiment

4.1 Experiment Setup

4.1.1 Datasets

I experiment with my proposed architecture on the Amazon Product Reviews dataset (Amazon), Web-of-Science[5] (WOS), and Wikipedia Categories(Wiki) datasets for comparison and ablation study.

The distribution of parent and child labels in the Amazon dataset is uneven, indicating a substantial variation in the number of review categories.

In the Wiki dataset, while the distribution of parent labels is balanced, the distribution of child labels under each parent label is uneven, suggesting that specific topics encompass a more significant number of subcategories.

In the WoS dataset, the child labels exhibit a relatively uniform distribution due to their wide variety. Still, the parent labels vary significantly, indicating that specific research fields contain far more articles than others.

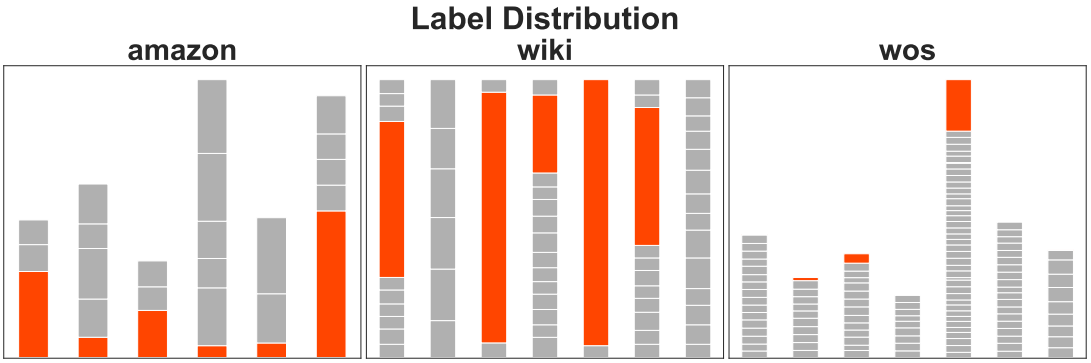


Fig. 4.1 The label distribution across different datasets.

4.1.2 Implementation Details

I utilize a one-layer bi-GRU with 64 hidden units, along with 3 parallel CNN layers with filter region sizes of $\{2, 3, 4\}$. The vocabulary is constructed from the most frequent words, with a maximum size of 60,000. I employ 300-dimensional pretrained word embeddings from GloVe[2] [10], while out-of-vocabulary words with a minimum count of 2 are randomly initialized. Since key information relevant to text classification often appears in the initial parts of the text, the maximum token input length is set to 256. A fixed threshold of 0.5 is used for tagging. Dropout is applied in the embedding and MLP layers at a rate of 0.5, while the rates are set to 0.1 and 0.05 for the bi-GRU layer and node transformation, respectively.

4.2 Experiment Results

Table 4.1 Comparison of experimental results in various datasets.

Dataset	HiAGM		**HiFRM**	
	Micro	Macro	Micro	Macro
WIKI	85.95	52.30	**85.79**	**52.65**
AMAZON	73.97	65.61	**74.62**	**66.08**
WOS	84.85	78.90	**85.92**	**79.66**

The experimental results in Table 4.1 demonstrate that the HiFRM model outperforms the HiAGM model on the WIKI, AMAZON, and WOS datasets in most cases. In terms of Micro-F1, HiFRM surpasses HiAGM on the AMAZON and WOS datasets, achieving scores of 74.62 and 85.92, respectively, while slightly trailing HiAGM on the WIKI dataset (85.79 vs. 85.95). For the Macro-F1 metric, HiFRM achieves consistent improvements across all three datasets, with scores of 52.65 on WIKI (compared to HiAGM's 52.30), 66.08 on AMAZON (compared to 65.61), and 79.66 on WOS (compared to 78.90). Overall, HiFRM demonstrates balanced performance across both Micro-F1 and Macro-F1 metrics, showcasing its superior effectiveness across diverse datasets.

In summary, the HiFRM model demonstrates outstanding performance in terms of the Micro-F1 metric, particularly surpassing HiAGM on the AMAZON and WOS datasets. Additionally, HiFRM achieves consistent advantages in the Macro-F1 metric

across all datasets. Overall, the results indicate that HiFRM presents a competitive architecture, delivering excellent performance across diverse datasets.

4.3 Ablation Study

Table 4.2 Ablation study on WOS dataset.

Ablation Models	Micro	Macro
HiFRM(FULL)	85.92	79.66
w/o Hierarchy-Feature Propagation	84.62	76.08
w/o Feature-Aware Matric	85.61	79.48

The ablation study on the WOS dataset, as shown in Table 4.2, demonstrates the independent contributions of the two main components in the HiFRM model: hierarchy-feature propagation and feature-aware matrix. When hierarchy-feature propagation is removed, the performance drops significantly, with Micro-F1 decreasing to 84.62 and Macro-F1 to 76.08. This is because hierarchy-feature propagation effectively captures hierarchical relationships in the data, and its absence leads to insufficient exploitation of hierarchical structures, particularly impacting the Macro-F1 score. Similarly, removing the feature-aware matrix results in a decline in performance, with Micro-F1 dropping slightly to 85.61 and Macro-F1 to 79.48. The feature-aware matrix plays a crucial role in enhancing feature interaction, and its removal limits the model's ability to leverage informative features, particularly for less frequent tables, which is reflected in the lower Macro-F1 score. These results highlight that both components are critical for the HiFRM model to achieve optimal performance.

4.4 Conclusion

In this research, I propose an end-to-end hierarchy-aware text feature propagation model that leverages fine-grained label-related information. The model incorporates a hierarchy-GCN as a text structure encoder and introduces an evaluation module based on global label features to provide a more comprehensive measure of effectiveness. Empirical results demonstrate that the proposed HiFRM model achieves significant improvements across three distinct datasets, with particularly strong performance on low-frequency labels.

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